

Point-by-Point Response to the Editor and Reviewers

Zhi Huang, on behalf of all authors

19 July 2026

Submission ID: 56f53c41-a86e-4948-8cfd-a3c11b83fd1b

Manuscript title: BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge

Dear Ms Agnihotri, Dr Zoete, and Reviewers,

We thank the Editor and all three reviewers for their detailed and constructive assessment. The revision follows a conservative evidence-first approach. We retain the original Davis records, pre-processing, 70/10/20 ratios, and seed 42, then add only reproducible diagnostic analyses that can be supported by the available data. In particular, we (i) audit duplicate sequences and full-length target identity; (ii) add exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations; (iii) add leakage-aware drug-only, drug-promiscuity, target-only, and label-shuffled controls; (iv) report performance by target-identity bins; and (v) narrow all claims about generalisation, structural interpretation, medicinal-chemistry utility, and resistance modelling. We do not represent unperformed external-dataset, structural-contact, or matched resistance experiments as completed.

Revision-location guide. Page references below use the compiled clean manuscript dated 19 July 2026. The Abstract and Introduction are on pp. 1–2; Related Work on pp. 2–3; Methods on pp. 3–6; Experimental Setup on pp. 6–7; Results on pp. 7–14; Discussion on pp. 15–16; Conclusion on p. 16; the web-server figure is on p. 17; and availability statements are on p. 18. In the Supporting Information, model/training and environment details are on pp. 1–2, split summaries are on p. 3, reproducibility details are on p. 4, input/case/parameter audits are on pp. 5–6, and strict BindingDB provenance is on p. 7. This guide supplements the section-level locations provided after each response.

Response to the Editor

E.1. Validation, overstatement, and manuscript/code reconciliation

Reviewer comment (full text). *“The reviewers have identified substantial limitations. In particular, the Methods and experimental evaluation should be strengthened by providing more rigorous validation, including broader and fairer benchmarking, better justification of dataset and baseline selection, more comprehensive ablation and interpretability analyses, and appropriate controls for the reported claims. It is also important to ensure that all conclusions are fully supported by the presented evidence, address potential sources of bias and confounding factors, and clarify any discrepancies between the manuscript and the accompanying implementation.”*

Response. We have addressed the parts that can be substantiated from the submitted Davis material and have explicitly identified the remaining evidence limits. The revised manuscript now (i) distinguishes the original target-identifier-held-out result from sequence-novel transfer; (ii) adds exact sequence grouping, exact-sequence-grouped cold-target and exact-sequence-grouped cold-both tests; (iii) quantifies drug-side signal and target relatedness with controls and identity-stratified results; (iv) restores the numerical baseline comparison and component-removal ablation results, updates the BioInteract full-model entries to the canonical current-release metrics, and limits the

resulting component differences to descriptive evidence; and (v) redefines attention and Grad-CAM as model-native attributions rather than structural contacts. The web figure and command-line statements now describe only functionality supported by the repository and displayed screenshot. We do not claim that the new within-Davis controls replace external benchmarking, fair matched-encoder baselines, or structural validation.

During final pre-submission reconciliation, we also found that the legacy entity-split prediction CSVs and result JSON files could not be regenerated from the released checkpoints, current inputs, and code. We therefore rebuilt the three entity-split artifacts by instantiating each checkpoint from its embedded configuration, collecting full-precision predictions, selecting each F1 threshold once on validation data, and applying that threshold unchanged to the test data. The canonical artifact records hashes for checkpoints, data inputs, cached embeddings, and model configurations; it now drives every reported entity-split number, prediction CSV, bootstrap interval, and performance figure. For the restored component-removal table, we now use those same canonical current-release values for the displayed full BioInteract row, while retaining the component-removed rows exactly as reported in the original submission. Figure 2 is recalculated from the displayed rows. This is a transparent numerical reconciliation, not a new controlled ablation: its row-to-row differences are described as component sensitivities and are not used as causal matched-retraining estimates.

We also completed a final manuscript-code reconciliation and consistency audit:

- Figure 3 now presents the canonical entity-split reconstruction as Panel A and the separately trained exact-sequence-grouped strict evaluations as Panel B; the identifier-based protocols are named “Target-ID-held-out” and “Drug-ID-held-out” consistently.
- The protein-side optional channel is named the “domain-label channel” or, where its actual content matters, the “unknown-label embedding channel.” We define “biological prior knowledge” as the frozen ESM-2 representation plus residue physicochemical features, not Pfam/InterPro annotation.
- The final checkpoint reconstruction uses each checkpoint’s embedded architecture configuration, direct full-precision inference, and a validation-selected frozen threshold. The updated BioInteract values are used in the numerical baseline-comparison table and figure.
- The original numerical component-removed rows are retained, while the displayed full BioInteract row is updated to the canonical current-release metrics. Figure 2 is regenerated from these displayed values, and the text explicitly limits its mixed-row differences to descriptive evidence rather than controlled component-effect estimates.
- The attention-sparsity Methods now give the normalisation and sparsity equations, checkpoint, sample scope, and residue-pooling rule. The unused arbitrary “hotspot” definition has been removed.
- The Methods/SI now report the alignment function and scoring parameters, normalised-Indel implementation, non-chiral Morgan setting, exact software versions, ordinary pair-bootstrap procedure, seed and percentile interval, and the Grad-CAM layer and pre-sigmoid target.

Revision in manuscript: Abstract; Introduction; Related Work; Methods, “Training Provenance and Evaluation Thresholds”, “Attention Sparsity”, “Grad-CAM Functional-Group Attribution”, and “Data Splitting Protocols and Similarity Audit”; Results, “Comparison with Baseline Methods” and “Ablation Study”; Discussion; Supporting Information Tables S3–S7 and S13; Figs. 1–7 and captions.

E.2. Archive code in a DOI-issuing repository

Reviewer comment (full text). “Please note that if your manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or a pipeline (even if individual components are not new), the underlying code must be deposited in a recognised DOI-assigning repository (e.g. Zenodo) and linked either from Methods or a dedicated Code Availability section.”

Response. We added a dedicated Code Availability section with direct links to the public source repository, <https://github.com/Liubo-520/bio-intelligence>, and the immutable Zenodo software archive, <https://doi.org/10.5281/zenodo.21437753>. The archive contains the released code, pretrained weights, configuration files, canonical prediction CSVs, provenance hashes, figure-generation scripts, strict BindingDB aggregate manifests and audit, and reproduction instructions; the Hugging Face Space provides the custom-prediction interface described in the revised manuscript.

Revision in manuscript: New “Code Availability” section; Data Availability section.

Response to Reviewer 1

General assessment and strengths

Reviewer comment (full text). General assessment

The manuscript presents BioInteract, an interpretable DTI prediction framework based on pharmacophore-aware molecular graphs, ESM-2 residue embeddings, domain annotations, and bidirectional atom-residue cross-attention. The work is technically interesting and the Davis benchmark results are encouraging, especially in the cold-target setting.

However, the main claims remain stronger than the current evidence supports. The model addresses a binary high-affinity classification task, whereas several sections interpret attention and Grad-CAM outputs as direct evidence of binding mechanisms. Additional controls are needed to distinguish genuine interaction learning from ligand-side QSAR, target homology, and activity-profile biases.

Strengths

The architecture is well motivated and combines relevant molecular and protein representations. The comparison includes random, cold-drug, and cold-target splits, which is appropriate for this type of task. The manuscript is generally clear, and the supplementary information provides useful implementation and split details.

Response. We appreciate this balanced assessment. The revision directly follows the distinction identified here: BioInteract is now described as a binary high-affinity classifier on the Davis benchmark, while attention and Grad-CAM are described as model-native attribution outputs rather than direct evidence of binding mechanisms. The strict split audit, ligand-side controls, target-relatedness analysis, and restricted case-study language address the stated risks of QSAR, homology, and activity-profile bias in the sections detailed below.

Revision in manuscript: Abstract; Introduction; Methods, Problem Formulation and Data Splitting Protocols and Similarity Audit; Results; Discussion and Conclusion.

R1.1. Simpler QSAR and dataset-bias explanations

Reviewer comment (full text). 1. Simpler QSAR and dataset-bias explanations are not excluded.

The ablation study does not sufficiently test whether BioInteract learns drug-target interaction patterns, as opposed to ligand-side activity profiles or matrix-completion effects. The w/o ESM-2 model still receives target information, and the w/o cross-attention model does not test how far a simple ligand-only model can go.

This is important because, in the cold-target split, the proteins are unseen but the ligands may already be present in training. The authors should include diagnostic baselines, especially a simple drug-only multilabel QSAR model predicting kinase activity profiles without protein input. Target-only, degree/promiscuity-based, and shuffled-pair controls would also help determine how much performance remains without explicit interaction modelling. If these controls perform well, the claims about learned molecular-recognition principles should be narrowed.

Response. We agree that the submitted ablation did not rule out drug-side signal. A target-profile multilabel predictor cannot be evaluated for an unseen target without introducing its held-

out labels; we therefore use a pair-level, drug-only Morgan logistic diagnostic that receives no target representation and only training-partition labels. We also state explicitly that the label-shuffled control is a label-sanity control rather than the requested structure-level shuffled-pair experiment. We therefore added four leakage-aware diagnostic controls to the same exact-sequence-grouped cold-target test (5,984 pairs; 208 positives). The drug-only Morgan logistic control, which receives no target representation, achieves AUROC/AUPRC of 0.764/0.164. A drug-promiscuity prior estimated only from training targets gives 0.764/0.163. In contrast, the only target-only control that is valid for unseen target identifiers is the uniform training prevalence (0.500/0.035): using a held-out target’s full label degree would leak test labels. A deterministic training-label-shuffled drug-only control gives 0.407/0.029. BioInteract achieves 0.873/0.383 on the same strict test.

These results show that ligand-side activity/promiscuity signal is material in Davis. BioInteract exceeds the diagnostic controls, but we do not interpret the difference as proof of physical interaction learning or structural recognition. The manuscript now states this boundary explicitly. We describe the shuffled-label test accurately as a label-sanity control, not as a structure-level shuffled-pair experiment.

Table 1: Evidence for R1.1: leakage-aware controls on the exact-sequence-grouped cold-target test.

Control or model	AUROC	AUPRC
BioInteract (exact-sequence-grouped cold-target)	0.873	0.383
Drug-only Morgan logistic	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only	0.407	0.029

Revision in manuscript: Methods, “Data Splitting Protocols and Similarity Audit”; Results, new Table “Leakage-aware diagnostic controls” and revised manuscript Fig. 4; Discussion, new “Diagnostic Controls and Remaining Dataset Bias”.

R1.2. Control cold-target performance for target homology

Reviewer comment (full text). 2. *Cold-target performance should be controlled for target homology.*

The cold-target result is strong, but its interpretation is currently too broad. Davis contains kinase inhibitors and kinase targets, so a test target can be absent from training while still being highly homologous to training kinases. The result may therefore reflect transfer within the kinome more than generalisation to biologically distant targets.

The authors should report the maximum sequence identity between each cold-target test protein and the training targets, preferably for both full-length sequences and kinase domains. A homology-reduced cold-target split, for example after clustering targets at a defined identity threshold such as 60%, would be particularly informative. At minimum, performance should be stratified by target-similarity bins.

Response. We added a full-length global-alignment audit for the exact-sequence-grouped cold-target split. For each of 88 held-out targets, we report the maximum exact-residue identity divided by aligned length against all 310 training targets. The maximum identity ranges from 0.231 to 0.867 (median 0.494). The four identity bins contain 31, 28, 24, and 5 targets below 0.40, 0.40–<0.60, 0.60–<0.80, and at least 0.80, respectively. Corresponding pair-level BioInteract AUROC/AUPRC values are 0.806/0.305, 0.908/0.456, 0.908/0.466, and 0.862/0.121. The final bin contains only three positives, so its AUPRC is unstable. These non-monotonic results make residual relatedness visible but do not establish distant-family transfer.

We did not claim a kinase-domain-specific identity analysis or a 60%-clustered split because no independently curated kinase-domain boundaries or new split results were supplied. Instead, the

manuscript now makes the full-length analysis and its limitation explicit.

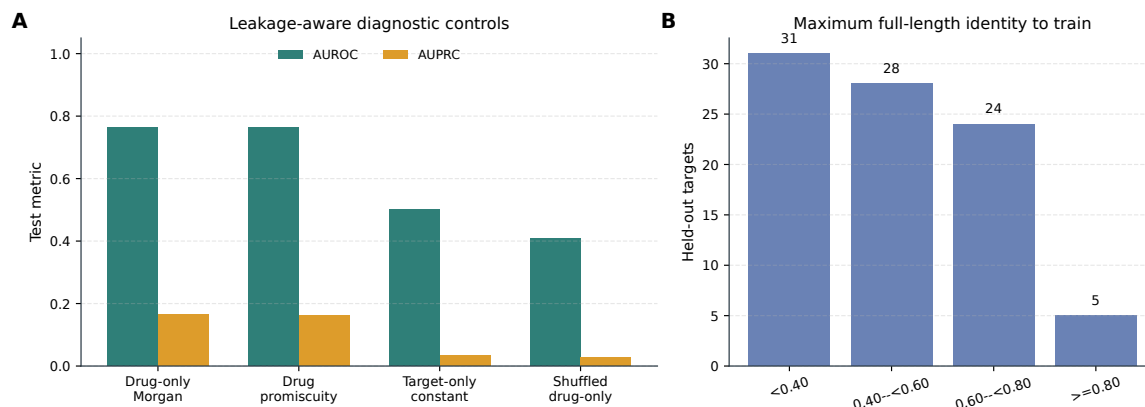


Figure 1: Evidence for R1.1–R1.2. Panel A shows leakage-aware controls; Panel B gives the count of exact-sequence-grouped cold-target test proteins by maximum full-length identity to a training target.

Revision in manuscript: Methods, “Data Splitting Protocols and Similarity Audit”; Results, new diagnostic-control table and revised manuscript Fig. 4; Discussion, “Scope of Cold-Start Evaluation”.

R1.3. Resistance-mutant analysis

Reviewer comment (full text). 3. *The resistance-mutant analysis does not yet show resistance-aware modelling.*

The ABL1 mutant examples mainly show conserved attention across selected mutant cases that remain high-affinity binders. This supports attention stability across similar positive cases, but it does not demonstrate that the model captures resistance-associated loss of activity.

The authors should include matched WT/mutant examples for the same inhibitor where the mutation experimentally reduces binding or activity. The model should then show both a lower predicted binding probability and an interpretable shift in the residue-level attention map. Without this analysis, the section should be reframed as attention conservation across selected ABL1 binder cases.

Response. We thank the reviewer for prompting a closer audit of the actual Davis model inputs. The nominal ABL1 wild-type, F317I, F317L, M351T, E255K, and suffix-labelled records used here have identical 1,167-residue sequences and the same SHA-256 digest in `target_sequences.csv`. Thus the model never receives a sequence change that corresponds to the nominal variant label. We have removed the entire previous mutant-comparison analysis, including its tables, figures, case study, and all resistance-related conclusions, rather than reframe it as a stability result. The revised manuscript states this limitation transparently, and Supporting Information Table S9 records every identifier, length, digest, and the associated D0010/D0017 Davis records.

Revision in manuscript: Results, “Audit of Nominal ABL1 Variant Inputs”; Discussion, “Limitation of Nominal ABL1 Variant Records”; Supporting Information, Table S9.

R1.4. Validation of interpretability analyses

Reviewer comment (full text). 4. *The interpretability analyses need stronger validation.*

The attention sparsity, Grad-CAM pharmacophore rankings, target-dependent pharmacophore profiles, and hotspot-count analysis are interesting, but they remain exploratory. Sparse attention does not by itself prove binding-pocket localisation, and Grad-CAM agreement with general kinase-inhibitor SAR is not sufficient mechanistic validation.

The authors should compare high-attention residues with experimentally resolved ligand–protein contacts when structures are available. They should also clarify how atom-level Grad-CAM scores are aggregated into functional groups and test whether these profiles are stable across related ligands or matched analogues. The BRAF example, which is a low-confidence false-negative case, should be treated as a failure-mode analysis rather than as positive evidence for target-dependent pharmacophore usage. The hotspot-count analysis should also be softened, since it is based on few selected cases and an arbitrary attention threshold.

Response. We agree that sparse attention and Grad-CAM are exploratory without independent supervision. The Methods now define functional-group values as aggregated atom-level Grad-CAM attributions and state that they are neither contacts nor interaction types. The manuscript cites structurally supervised work such as MONN to distinguish that setting from the present one. No PDB/PLIP contact comparison or matched-analogue stability test was performed, and none is claimed. The BRAF/D0023 result is now a low-confidence failure-mode example, not positive evidence for target-dependent functional-group usage. We removed the hotspot-count analysis and figure because the selected cases included nominal ABL1 variants with identical model-input sequences and the arbitrary threshold could not support a structural or affinity claim.

For reproducibility, the revised Methods define residue score $s_j = \sum_i M_{ij}$, pair-specific normalisation $\tilde{s}_j = s_j / \max_k s_k$, and Sparsity = $|\{j : \tilde{s}_j < 0.1\}|/L$. The reported 99.2% value uses checkpoints/best.pt and all 1,506 positive Davis pairs, including original training, validation, and test memberships. Each pair is normalised before pooling all 1,301,380 residue observations; consequently, longer proteins contribute more residue-level observations. Grad-CAM is computed at the final (third) GINE layer with respect to the pre-sigmoid interaction logit. No residue is designated a biological hotspot.

Revision in manuscript: Related Work, “Interpretability in DTI Models”; Methods, “Grad-CAM Functional-Group Attribution”; Results, non-mutant interpretability case studies; Discussion, “Functional-Group Attribution Without Structural Supervision”; removed former hotspot-count analysis and figure.

R1.5. Figure 2 readability

Reviewer comment (full text). Minor comment: Figure 2 should be redrawn for readability, especially the right panel.

Response. We restored the component-removal numerical rows and redrew the figure for readability. The revised Figure 2 now identifies Panels A and B, separates the metric heatmap from the AUROC-difference panel, moves the legend below the bars, and uses non-overlapping labels. Panel B now includes all four component-removed rows, including graph augmentation, so the figure and Table 2 carry the same information. To make the numerical reconciliation explicit, the displayed full BioInteract row uses the current canonical entity-split metrics; the component-removed rows retain their original submitted values. Accordingly, Panel B is labelled as a descriptive difference from the displayed full-model row, not as a matched-retraining causal-effect plot.

Revision in manuscript: Results, “Ablation Study”; Table 2 and Figure 2.

R1.6. Training dynamics and overfitting

Reviewer comment (full text). Minor comment: Section 5.4 should avoid stating that the training curves confirm absence of overfitting. Validation AUPRC, validation loss, and repeated-seed variability would make this point more convincing.

Response. We agree. We removed the assertion that single-seed training curves confirm absence of overfitting. Following final log audit, Figure 5 was rebuilt from one contiguous trace per protocol whose peak validation AUROC matches its saved checkpoint; duplicated and restarted log segments were excluded. The revised text limits the figure to observed validation-AUROC and training-loss trajectories and explicitly states that they do not quantify validation loss or repeated-seed variability.

Revision in manuscript: Results, “Training Dynamics” and Figure 5 caption.

R1.7. Binary classification and cautious cold-drug interpretation

Reviewer comment (full text). Minor comments

The manuscript should consistently distinguish binary high-affinity classification from quantitative affinity prediction. Claims related to potency, lead optimisation, and mechanistic binding should be moderated accordingly.

The cold-drug results should be discussed more cautiously. BioInteract improves over the selected baselines, but the absolute AUPRC and thresholded metrics indicate limited generalisation to unseen molecules.

Response. We revised the Introduction, Methods, Results, Discussion, and Conclusion to define the task as binary classification of Davis pairs at $K_d < 30$ nM. We removed potency, lead-optimisation, and ranking language from the present results. Continuous affinity prediction is stated only as future work. We also clarify that Drug-ID-held-out means unseen identifiers, not validated scaffold hopping or out-of-distribution chemical generalisation; its AUPRC and thresholded metrics are interpreted cautiously.

Revision in manuscript: Abstract; Introduction; Methods, “Problem Formulation” and “Evaluation Metrics”; Results, Drug-ID-held-out discussion; Discussion and Conclusion.

Reviewer 1 recommendation

Reviewer comment (full text). Recommendation: Major revision.

The study is promising, but the central claims require stronger diagnostic controls and more cautious interpretation. The key revisions are a ligand-only QSAR control, a homology-aware cold-target evaluation, a more informative resistance-mutant analysis, and structural or SAR-based validation of the interpretability outputs.

Response. We appreciate this clear recommendation. The added ligand-only and promiscuity controls, exact-sequence-grouped cold-target evaluation, and full-length identity stratification directly address the first two requests. As the nominal ABL1 variants have identical model-input sequences, the mutant analysis was withdrawn in full. Because matched loss-of-activity and structural-contact data were not available, the remaining non-mutant attribution analyses are explicitly presented as exploratory model behaviour rather than validated mechanistic findings.

Revision in manuscript: Results and Discussion revisions identified in R1.1–R1.7.

Response to Reviewer 2

R2.1. Methodological novelty

Reviewer comment (full text). 1. *The combination of ESM-based protein representations and graph-based molecular encoders, often together with attention-based interaction modules, is not entirely new in DTI/DTA prediction. Therefore, the manuscript should more clearly clarify the methodological novelty of the proposed framework compared with existing ESM + graph encoder models and their attention-based extensions.*

1) PGraphDTA: Improving Drug Target Interaction Prediction using Protein Language Models and Contact Maps

2) HCAF-DTA: drug-target binding affinity prediction with cross-attention fused hypergraph neural networks

3) GEFA: Early Fusion Approach in Drug-Target Affinity Prediction

etc.

Response. We agree that these individual ingredients are not new. The revised Related Work now explicitly cites GEFA, PGraphDTA, and HCAF-DTA and states that BioInteract does not claim first use of a protein language model, a graph encoder, or cross-attention in DTI/DTA prediction. We have narrowed the claimed contribution to the submitted integration of chemistry-aware molecular graphs, residue-level ESM-2/physicochemical features with a configured unknown-label embedding channel, and model-native attribution outputs for Davis binary classification. The new text also cites 3DProtDTA, the domain-specific framework of Liu *et al.*, and InceptionDTA when discussing the broader multi-benchmark DTA literature requested by the reviewers. These citations are used to situate the scope and do not imply a direct numerical comparison across different task definitions, label transformations, data processing, or splits. Finally, we clarify throughout that the attribution outputs are not independently validated structural evidence.

Revision in manuscript: Related Work, “Pretrained Protein Language Models” and “Interpretability in DTI Models”; Introduction and Conclusion.

R2.2. Fair comparison with pretrained protein representations

Reviewer comment (full text). 2. *The cold-start comparison may not be entirely fair because BioInteract uses frozen ESM-2 embeddings pretrained on a large-scale protein sequence corpus, whereas the baseline models included in the manuscript do not appear to use comparable pretrained protein representations. Therefore, the reported improvement, especially under the cold-target split, may largely reflect the benefit of ESM-2 pretraining rather than the proposed architecture itself. The authors should include stronger baselines that also use pretrained protein representations, such as ESM-based DTI/DTA models, or provide controlled comparisons where the same pretrained protein features are used across competing architectures.*

Response. We agree that the baseline comparison does not isolate the effect of ESM-2 pretraining. We restored the numerical baseline comparison but revised its interpretation: it is presented as Davis benchmark context, not as controlled evidence that the cross-attention architecture alone produces the observed differences. The current reconstructed BioInteract entity-split values are used in Table 1 and Figure 1, and the text explicitly states that no matched ESM-based baseline was newly evaluated.

For the restored component-removal values, the full BioInteract row now uses the same canonical current-release entity-split metrics that appear in Table 1 and Figure 1, while the component-removed rows retain their original submitted numbers. We regenerate Figure 2 from the resulting displayed table, disclose the mixed provenance in its caption and text, and do not interpret the differences as controlled causal effects of ESM-2, cross-attention, domain labels, or graph augmentation. This avoids both an undocumented numerical inconsistency and an unsupported attribution of performance to a single architectural component.

Revision in manuscript: Results, “Comparison with Baseline Methods” and “Ablation Study”; Tables 1–2; Figures 1–2; Discussion.

R2.3. Davis-only evaluation and dataset bias

Reviewer comment (full text). 3. *The Davis dataset is highly kinase-specific and is known to have strong dataset bias. Evaluating the model only on this dataset makes it difficult to objectively assess its general DTI prediction performance. I recommend additional evaluation on larger and more diverse datasets, such as BindingDB, which contains a broader range of proteins and drug-target interactions.*

Response. We agree and added a pre-specified frozen external evaluation using the official BindingDB curated-articles snapshot `BindingDB_BindingDB_Articles_202607`. We retained only exact, finite K_d measurements, used the same $K_d < 30$ nM label definition, excluded invalid or multi-chain records, and then excluded every ligand whose canonical isomeric SMILES or target whose full normalised sequence occurred anywhere in Davis. The final strict double-novel cohort contains 1,932

pairs (516 positive and 1,416 negative) from 1,195 ligands and 396 targets.

For complete curation accounting, the sequential row-level filters leave 2,158 eligible measurements. The 15 threshold-conflicting ligand–target groups are counted at the pair level and comprise 44 measurement rows; 2,114 threshold-consistent rows remain. For each consistent pair, we retain the median exact K_d ; 182 repeated measurement rows across 143 retained pairs are thereby collapsed, yielding 1,932 final pairs. This procedure is now specified in the Methods and Table S13.

The released `best_random` checkpoint was evaluated without retraining, fine-tuning, recalibration, checkpoint selection, or threshold tuning on BindingDB. At the frozen Davis random-validation threshold, it obtained AUROC = 0.560 (95% CI 0.531–0.589) and AUPRC = 0.340 (95% CI 0.314–0.372); F1, precision, and recall were 0.070, 0.731, and 0.037. The result does not support a broad DTI-transfer claim. Rather, it documents limited transfer of the frozen Davis-trained classifier to this broader exact-entity cohort. It leaves the within-Davis checkpoint reconstructions unchanged and is not used as evidence for the attention, attribution, structural, or mechanistic analyses.

Revision in manuscript: Methods, “Pre-specified External BindingDB Evaluation”; Results, “Strict External BindingDB Evaluation” and Table 5; Supporting Information, Table S13; Discussion and Conclusion.

R2.4. Baseline selection and recency

Reviewer comment (full text). 4. The baseline selection also appears somewhat limited and outdated. Among the compared methods, the most recent baseline appears to be DrugBAN, which is marked as a 2023 model in the manuscript. Since more recent DTI/DTA models have been proposed after 2023, including models using graph-based molecular encoders, attention-based fusion, and pretrained protein representations, the authors should include stronger and more up-to-date baselines.

Response. We now cite verified examples of recent PLM, graph, and attention models. We restored the original numerical comparison, but it is presented as benchmark context rather than a current state-of-the-art leaderboard or a controlled architecture comparison. We do not claim to have reproduced additional baselines without matching their task definition, preprocessing, and pretraining. A fair comparison requires a common protocol and matched pretrained target features, which remain outside the evidence available in this revision.

Revision in manuscript: Related Work; Results, “Comparison with Baseline Methods”; Discussion, “Future Directions”.

R2.5. Biological meaning of attention maps

Reviewer comment (full text). 5. The attention map represents the degree to which the model attends to protein residues or drug atoms. However, Figure 5 alone is insufficient to conclude that the model has truly identified biologically meaningful interaction regions. Previous studies such as BlendNet and MONN validated attention maps or pairwise interaction prediction maps using tools such as PLIP to compare predicted interaction regions with actual protein–ligand contacts. A similar validation would be necessary to support the interpretability claim.

- 1) Exploring the potential of compound–protein complex structure-free models in virtual screening using BlendNet
- 2) MONN: A Multi-objective Neural Network for Predicting Compound-Protein Interactions and Affinities

Response. We agree. We removed language that treated attention maps as identified biological interaction regions. The revised text distinguishes the present unsupervised attribution output from models trained or evaluated with explicit non-covalent interaction labels. We did not run PLIP/PDBBind or BlendNet/MONN-style structural validation, so the response and manuscript state that a non-overlapping structural benchmark and defined contact protocol are required before assigning structural meaning.

Revision in manuscript: Related Work, “Interpretability in DTI Models”; Results, “Case Study: Functional-Group Attribution” and “Cross-Case Functional-Group Attribution Variation”; Discussion, “Functional-Group Attribution Without Structural Supervision” and “Future Directions”.

R2.6. Mutation-pattern analysis

Reviewer comment (full text). 6. It is unclear why analyzing mutation patterns is important for this model. If the goal is to demonstrate interpretability, simply analyzing residues with high attention scores is not sufficient. More detailed biological or structural analysis is needed to show that the model captures meaningful mechanisms rather than merely producing attention weights.

Response. We agree. A direct audit showed that all nominal ABL1 variant inputs used by the model have an identical amino-acid sequence. We therefore removed the former mutation-pattern analysis in full and do not use it as evidence of a biological mechanism. The revised Results and Discussion state that these Davis records cannot support a mutation-effect or resistance analysis for BioInteract; the reproducible input audit is supplied in Table S9.

Revision in manuscript: Results, “Audit of Nominal ABL1 Variant Inputs”; Discussion, “Limitation of Nominal ABL1 Variant Records”; Supporting Information, Table S9.

R2.7. Remaining case studies

Reviewer comment (full text). 7. The remaining case studies would be more convincing only after the model has been validated on larger and more diverse datasets. Based on the current results, it is difficult to determine whether the proposed model truly provides reliable interpretability or mechanistic insight.

Response. We agree that the cases are illustrative rather than validation. The new BindingDB test supplies a strict external performance check, but its limited transfer result is not evidence for case-level interpretability or mechanism. All case-study language therefore describes observed model behaviour or a failure-mode example, and the Conclusion does not use the cases to establish a mechanistic conclusion. Structural validation remains necessary.

Revision in manuscript: Results, “Strict External BindingDB Evaluation” and all interpretability case studies; Discussion; Conclusion.

Reviewer 2 overall assessment

Reviewer comment (full text). Overall, while the proposed model is interesting, the current manuscript would benefit from clearer novelty claims, fairer baseline comparisons, evaluation on larger-scale datasets, and more rigorous validation of the interpretability results.

Response. We agree with these priorities. The revision adds a strict external BindingDB evaluation, reports its limited transfer transparently, narrows the novelty claim, labels baseline comparison limitations, and removes structural or mechanistic interpretation claims that are not independently validated.

Revision in manuscript: Related Work; Methods, external evaluation; Results, baseline comparison, split audit, and external test; Discussion; Conclusion.

Response to Reviewer 3

R3.0. General comment: website and command-line accessibility

Reviewer comment (full text). General comments.

Although this work appears to be comprehensive, scientifically accurate, and technically sound, the authors should be careful to not overstate the capabilities of their framework, given the limitations of their dataset and the lack of explicit structural cross-validation (see questions below).

The authors have provided a codebase that is maintained on GitHub, facilitating the reproducibility of the findings and data analysis performed in this paper. However, the publicly accessible huggingface.co website does not appear to work properly for a couple of the pre-computed case studies (BRAF + Drug 11717001 (Sorafenib analogue) and EGFR + Drug 156414). Namely, the Top Predicted Binding Residues and Pharmacophore Group Importance (Grad-CAM) plots are not displayed for these case studies, nor are the predicted binding probability, model confidence, or experimental affinity shown. These plots also seem to be missing from the corresponding GitHub folder (the hf_space subfolder) in the repository, although Grad-CAM molecule figures seem to be available for 156414-EGFR and 11717001-BRAF (inside the results/interpretability/gradcam/ folder in the repository). Reassuringly, the Custom Prediction tab appears to be working for a user-provided drug SMILES and protein amino acid sequence. The authors appear to omit some of the functionality provided in their repository codebase within their publicly available huggingface.co space, and it is not clear why some features appear to be available within the code but not described within the context of the manuscript (see questions below). It would be useful if the authors could provide instructions on how to utilize the functionality of their GitHub repository to make predictions and generate outputs for custom ligands and targets through command-line operation, which would greatly increase the accessibility of the code and allow the larger scientific community to utilize the resource without having to input potentially proprietary information into a web server.

Response. We agree that the browser demonstration and the repository must not be conflated. Figure 7 is described as the custom-prediction input/result state shown for human Aurora kinase C (UniProt Q9UQB9), including the generated atom-residue attention map and “Top 10 Residue Attributions” chart visible in the lower portion of the screenshot. These are identified as model-native attribution views, not a residue-contact map, hotspot label, or experimental affinity. We do not use the displayed web case as evidence for the manuscript’s scientific claims.

For reproducibility, the revised Code Availability text directs readers to the repository README and identifies the supplied commands for training, evaluation, and archived-case attribution:

```
python -m src.cli.train
--config configs/default.yaml
python -m src.cli.evaluate
--config configs/default.yaml --checkpoint checkpoints/best.pt
python -m src.cli.interpret
--config configs/default.yaml --checkpoint checkpoints/best.pt
```

The current command-line interface does not constitute a general structure-based custom-ligand/pocket workflow, and no such functionality is claimed. This separates browser custom prediction from repository-based evaluation and attribution generation.

Revision in manuscript: Data Availability; Figure 7 caption; Conclusion.

R3.1. Rationale for Davis only and external validation

Reviewer comment (full text). 1. Can the authors comment further on their rationale for only choosing the Davis kinase inhibitor dataset to showcase the functionality of BioInteract? Was this done solely to facilitate comparison to existing drug-target interaction (DTI) software? By their own admission, this dataset is highly imbalanced (5% positive rate) and narrowly focused, comprising only 68 kinase inhibitor drugs and 442 protein kinase targets. Would the predictive ability of the model against diverse binding pockets and out-of-distribution chemical space not have been better showcased with a different dataset, such as BindingDB or KIBA (as mentioned in Section 6.6 – Future Directions), especially considering that the authors chose to combine ESM-2 embeddings with Pfam/InterPro domain information in their architecture? As it is currently presented, it is difficult to validate the claim that the BioInteract architecture is designed to be protein-family agnostic (which appears reasonable in theory) based only on the results obtained from the Davis dataset. A quick survey of other papers in the literature (e.g., doi: 10.1039/d3ra00281k, 10.1021/acs.jcim.4c00403, 10.1016/j.heliyon.2025.e42476) also shows that

other authors usually evaluate their model on multiple datasets in addition to the Davis dataset, including the KIBA, BindingDB, BIOSNAP, and PDBind datasets. While the authors mention analyses on other datasets as a direction for future research, they are still strongly encouraged to demonstrate the functionality of their model on at least one other such dataset, both to bring their work in line with others in the literature and to better assess the ability of their model.

Response. We agree that Davis-only evidence cannot establish broad protein-family, binding-pocket, or chemical-space generalisation. Davis was retained because it is the original submitted benchmark; it was not selected as a demonstration of protein-family-agnostic performance. All such language has been removed. We now add a pre-specified frozen BindingDB external evaluation rather than treating the within-Davis controls as external validation. The official curated-articles snapshot was filtered to exact finite K_d measurements and strict double novelty against all Davis ligands and full target sequences. The sequential row-level filters leave 2,158 eligible measurements. The 15 threshold-conflicting groups are pair-level counts and comprise 44 measurement rows; the 2,114 remaining rows are aggregated by the within-pair median exact K_d , collapsing 182 repeated rows to the final 1,932 pairs (516 positive and 1,416 negative). Without retraining, recalibration, or BindingDB threshold tuning, the released `best_random` checkpoint produced AUROC = 0.560 (95% CI 0.531–0.589) and AUPRC = 0.340 (95% CI 0.314–0.372). The result does not support a broad DTI-transfer claim. It instead documents limited transfer to a broader exact-entity cohort and is not used to support claims about binding pockets, attention maps, or protein-family-agnostic performance.

During final reproducibility auditing, we also found that the released package contains no per-protein domain-annotation JSON files. The archived loader assigns its `NONE` domain label when such a file is absent, so the released Davis inputs do not provide curated Pfam/InterPro annotations. We revised the abstract, Methods, ablation interpretation, and Supporting Information to describe a configured unknown-label embedding channel rather than a domain-enriched biological input. This correction avoids attributing either performance or interpretability to annotations that are not present in the reproducible assets.

We also added the three literature examples named in this comment—3DProtDTA, the domain-specific DTI framework of Liu *et al.*, and InceptionDTA—to the Related Work. They provide appropriate context for the reviewer’s request because they use broader benchmark coverage, but we do not present their reported scores as direct baselines: their endpoints, label definitions, preprocessing, and training protocols are not matched to the Davis binary-classification setup. The strict frozen-checkpoint BindingDB result is therefore retained as the external evidence for this revision, with its limited transfer stated plainly.

Revision in manuscript: Methods, “Pre-specified External BindingDB Evaluation” and “Configured Domain-Label Channel and Availability”; Results, external evaluation and ablation interpretation; Supporting Information, Table S13; Discussion and Conclusion.

R3.2. Cold-drug/cold-target definitions, similarity, and redundancy

Reviewer comment (full text). 2. *Can the authors elaborate further on the splitting method for their Cold-drug split and Cold-target split protocols? What is the Tanimoto similarity between the drugs in the training and testing sets for the Cold-drug split? Can the authors provide a pairwise similarity matrix or, for example, a t-SNE projection of the Morgan fingerprints of both sets? How do the authors define structurally novel molecules in this regard? Similarly, for the Cold-target split, what is the maximum sequence similarity of the test set proteins relative to the training set? It is extremely difficult to obtain proteins that are truly sequentially novel as the authors claim. From other publications (e.g., doi: 10.1016/j.heliyon.2025.e42476), it appears that the Davis dataset may contain up to 15% redundant protein sequences resulting from multiple identifiers per sequence. It is not evident that the authors have accounted for this fact.*

Response. We define the original Drug-ID-held-out protocol as drug-identifier exclusion, not a scaffold threshold or a claim of structural novelty. Radius-2, 1024-bit Morgan fingerprints were gen-

erated without chirality and give maximum test-to-training Tanimoto values of 0.196–0.538 (median 0.257) for the 13 held-out drugs. We report this reproducible test-to-training summary rather than presenting a t-SNE projection or a pairwise-similarity matrix as a definition of novelty.

For the original Target-ID-held-out split, the maximum RapidFuzz normalised Indel similarities range from 0.382 to 1.000 (median 0.579), and the audit finds 18 duplicate-sequence groups involving 81 target identifiers, with six groups crossing the original train/test boundary. We label that protocol Target-ID-held-out. We additionally provide the stricter full-length global-identity audit in R1.2; no sequence-novel claim is made.

Revision in manuscript: Methods, “Data Splitting Protocols and Similarity Audit”; Results, split-audit text and Fig. 4.

R3.3. Cold-both evaluation

Reviewer comment (full text). 3. *Related to Question 2, it appears that the authors have written code for the ability to perform a Cold-both split (inside `src/data/split.py`), in which neither ligand nor protein have been seen in the training data. As written in the comment of the function in the codebase, this is a true test of generalizability, and such functionality is essential for cases in which both the target and the molecules that could bind the target are unknown. Why do the authors choose to omit the description of this splitting approach in the main text, and why do they not report any comparative metrics using this splitting method? Such results could potentially augment the findings of the manuscript and provide a stronger case for the true power of the BioInteract attention framework.*

Response. We agree that the omission was not justified. We now report an exact-sequence-grouped cold-both test using the original records, preprocessing, architecture, and seed. Unlike the historical identifier-split reconstructions, each strict-split model was trained from initialisation on its corresponding new training partition. Both use weighted BCE with 0.05 label smoothing, AdamW (10^{-4} learning rate; 10^{-5} weight decay), batch size 64, five-epoch warm-up plus cosine decay, and validation-AUROC selection with patience 15. The complete artifact paths, SHA-256 hashes, selected epochs, and frozen validation thresholds are reported in Supporting Information Table S3. The 70/10/20 ratios are applied independently to drug identifiers and exact target-sequence groups; only pairs in the corresponding train–train, validation–validation, and test–test blocks are retained. This produces 15,190/264/1,144 train/validation/test pairs and excludes 13,458 cross-block pairs, so 70/10/20 is not a pair-level proportion for this protocol. Exact-sequence-grouped cold-target gives 0.873/0.383 AUROC/AUPRC; exact-sequence-grouped cold-both gives 0.552/0.038 with 38 positive test pairs. The cold-both validation block has only 10 positives, and at its validation-selected threshold (0.936), test F1, precision, and recall are all 0.000. Both protocols are labelled within-Davis stress tests rather than proof of broad generalisation.

Table 2: Evidence for R3.3: exact-sequence-grouped cold-start evaluations.

Protocol	Test pairs	Positives	AUROC	AUPRC	F1	Recall
Exact-sequence-grouped cold-target	5,984	208	0.873	0.383	0.468	0.428
Exact-sequence-grouped cold-both	1,144	38	0.552	0.038	0.000	0.000

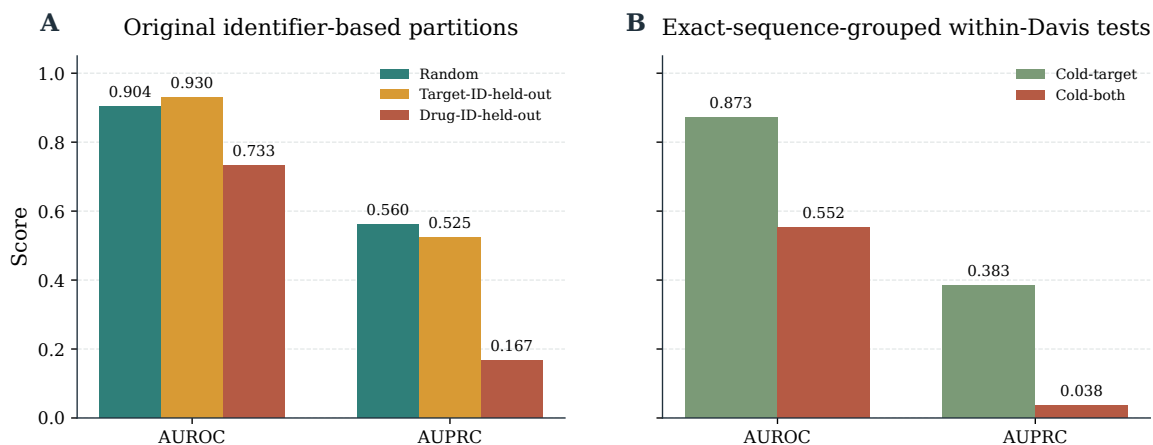


Figure 2: Evidence for R3.3: canonical current-release identifier-based splits (Panel A) and added exact-sequence-grouped strict splits (Panel B).

Revision in manuscript: Methods, split protocols; Results, Table “Strict within-Davis cold-start evaluation” and revised Fig. 3.

R3.4. Scope of interpretation and medicinal-chemistry claims

Reviewer comment (full text). 4. The authors make strong claims about the scientific insight gained by the BioInteract model, claiming in Section 6.1 that a medicinal chemist can use BioInteract not only to rank candidates but also to identify critical functional groups, map binding pockets, and assess potential resistance liabilities. However, the model does not provide any structural output, nor does it compute any binding scores or energies for ranking purposes. Moreover, there does not appear to be any way to directly visualize a labeled 2D-representation of the molecule (i.e., with atom labels to identify which atoms are predicted to interact), see the predicted binding pose of a putative small molecule binder in the binding pocket of the target protein (although functionality appears to exist inside `src/interpret/visualize.py` in the repository to generate a PyMOL visualization script), or simultaneously rank a large library of molecules against a target by a quantitative metric (as would be done in a virtual screen). Such functionality appears to fall outside the scope of the model. Structure-based design is a hallmark of medicinal chemistry and virtual screening approaches, and it is not directly addressed by this model. Although useful quantitative plots are provided through the residue interaction heatmaps and Grad-CAM scores, it is furthermore not clear how users can generate Grad-CAM molecule heatmaps through the `huggingface.co` website, or through the command line, as the authors have shown in Figure 9A. The authors point out that residue-level mechanistic explanations are furnished, but in reality, even residue-level is too coarse-grained when it comes to molecular interactions. Which atoms of the residue are interacting with which atoms of the ligand? Is it a residue sidechain interaction, or a residue backbone interaction? Is there a π -stacking interaction, cation- π interaction, van der Waals contact, H-bond, or polar interaction occurring with the ligand, and if so, which atoms of the protein residue are involved? Are any residues acting as H-bond donors or acceptors, and if so, which atoms of those residues are involved in the interaction? What is the environment of the binding pocket? These are the kinds of structure-based mechanistic explanations that drive medicinal chemistry and drug discovery campaigns. The authors should try to rephrase their language in the text to avoid making claims that are not fully supported by their approach, especially when they acknowledge in their Future Directions (Section 6.6) that direct structural validation is missing and would provide quantitative support for interpretability claims.

Response. We agree. Attention matrices and Grad-CAM are now called model-native attributions. We removed claims of binding-pocket mapping, interaction typing, atom-atom contacts, pose prediction, energy calculation, large-library ranking, and treatment-resistance assessment. The revised Grad-CAM text defines functional-group aggregation from atom-level attribution and reports saved non-mutant EGFR/BRAF summaries in the two functional-group attribution tables in Results; it

does not present an atom-contact or pose figure.

The repository README is identified as the route to the available training, evaluation, and archived-case attribution workflows. We do not claim that the present command-line interface provides a general custom-ligand Grad-CAM, PyMOL-pose, or structure-based interaction workflow. Those functions would require a separately validated structural protocol and are explicitly outside the scope of the present binary classifier.

Revision in manuscript: Abstract; Methods; Results, functional-group attribution tables; Discussion; revised web-interface figure and caption.

R3.5. Attention sparsity and BRAF failure mode

Reviewer comment (full text). 5. The authors’ reported observation regarding attention sparsity (Section 5.5.1 and 6.2), combined with the interesting failure mode reported for D0023 and BRAF (Section 5.8 and Fig. 10), raises an intriguing question about the training of BioInteract and the dataset used in this work. As the authors note, D0023 does not make strong H-bonding interactions, but rather interactions involving its aromatic ring, and has a low-confidence prediction despite its sub-nanomolar experimental binding affinity. How do we know that the model has not simply learned to associate H-bonding interactions to canonical kinase hinge-region residues as a predictor of binding affinity? As the authors themselves note in Section 6.3, the learned cross-attention patterns transfer via conserved binding-site geometry, implying that the cross-attention module would also be implicated in attending to molecules which do not make such canonical geometric interactions. Instead of classifying this as an out-of-distribution example, would this not rather fall under the category of a few-shot example that the model did not handle accurately due to dataset imbalances that influenced its attention mechanism? In this sense, can the authors also comment on the possibility that attention sparsity, rather than being an emergent property of the model, arises due to the limited and imbalanced nature of the training data itself?

Response. We agree that no single explanation is supported by the available analysis. The 99.2% value is now described only as a thresholded attribution concentration statistic that may reflect learned regularities, class imbalance, data composition, or their combination. BRAF/D0023 is now labelled a low-confidence failure-mode example. We do not attribute it specifically to out-of-distribution status, few-shot coverage, hydrogen-bond bias, or sparsity.

Revision in manuscript: Results, attention-sparsity and functional-group sections; Discussion, “The 99.2% Attention Sparsity”; Figure 6 and functional-group tables.

R3.6. Scalability and implementation version

Reviewer comment (full text). 6. Can the authors comment on the scalability of their model with larger datasets? Even with the current Davis dataset, containing only 68 unique drugs, the model comprises over 2.4 million trainable parameters, the majority originating from the 3-layer GINE drug encoder module. With a larger dataset, such as BindingDB, containing more than 1 million unique molecular entries, how feasible would it be to train the model? It seems like the training process only uses a single NVIDIA GPU, but this computational overhead would not necessarily be expected to scale linearly with data size. Also, for reproducibility, could the authors mention which version of Python was used in their implementation details? Although the requirements.txt file in the GitHub repository suggests that Python ≥ 3.8 was used, the authors do not explicitly state the version that they used in the text.

Response. The Methods now state Python 3.10.14, PyTorch/PyTorch Geometric, automatic mixed precision, and single-NVIDIA-GPU training. The model has 2,442,083 trainable parameters. We do not claim linear scalability to BindingDB-scale data; larger datasets require profiling of graph batching, embedding storage/extraction, optimisation time, memory, and potentially distributed training.

Revision in manuscript: Methods, “Implementation”; Discussion, “Future Directions”; Supporting Information parameter summary.

R3.7. Structural cross-validation

Reviewer comment (full text). 7. Have the authors considered cross-validation of their model with actual structural data, e.g., provided in the Leak-Proof PDBBind (LP-PDBBind) dataset, available on GitHub? As the authors acknowledge in their Future Directions section, direct structural validation will provide quantitative support for interpretability claims and illuminate attention patterns that agree with experimentally determined interactions. Have the authors checked to see whether any of the predictions in the Davis dataset overlap with available structural data in PDBBind? If so, can the authors report on any cases where the cross-attention metrics of BioInteract agree with established molecular interactions from a structural complex in PDBBind? If not, do the authors plan to conduct such cross-validation studies in their future work?

Response. No PDBBind or LP-PDBBind contact validation, including an overlap analysis between Davis cases and PDBBind complexes, was performed in the submitted study. We therefore report no structural agreement cases. The manuscript states that a non-overlapping structural benchmark, experimentally resolved complexes, and a predefined contact-comparison protocol are required before structural meaning can be assigned to the attributions. This cross-validation is identified as future work rather than represented as completed.

Revision in manuscript: Related Work; Results, interpretability text; Discussion, “Functional-Group Attribution Without Structural Supervision” and “Future Directions”.

R3.8. Allosteric and cryptic pockets

Reviewer comment (full text). 8. Not all interactions between drug molecules and proteins occur in canonical binding pockets with established catalytic residues. Allosteric binding interactions (e.g., those catalogued in the Allosteric Database [ASD], see e.g., doi: 10.1016/j.patter.2021.100408, doi: 10.64898/2026.03.21.713412) and cryptic binding pockets (see e.g., doi: 10.1016/j.sbi.2025.103215) greatly expand the druggable potential of the proteome. Moreover, cryptic pockets are often only revealed by protein dynamics, although recent work (e.g., doi: 10.1093/bioinformatics/btae745) also points to the potential of pretrained protein language models to implicitly encode such information. Could the authors comment on the robustness of the BioInteract model in the prediction of such non-canonical drug-target interactions that may not otherwise be captured by other DTI models? Do the authors plan to evaluate their model against allosteric or cryptic site benchmark datasets in their future work?

Response. The kinase-only Davis data neither annotate allosteric sites nor cryptic pockets, and the present model is not trained with site labels, conformational ensembles, or experimentally resolved contacts. We therefore make no robustness claim for non-canonical interactions. The revised Future Directions now cites the allosteric-site benchmarking perspective of Wu *et al.*, the recent allosteric-grammar preprint identified by the reviewer, the review of physics/AI approaches to cryptic pockets, and CryptoBench. These references make the distinction clear: a suitable future assessment must use pre-specified allosteric or cryptic-site labels, structural or dynamic context, and a non-overlapping evaluation protocol. We identify CryptoBench as one potential cryptic protein-ligand benchmark, but we do not imply that BioInteract has been evaluated on it in this revision.

Revision in manuscript: Discussion, “Future Directions”.

R3.9. Minor comment 1: abbreviation definitions

Reviewer comment (full text). Minor comment 1. The authors should make sure to define abbreviations for clarity. Several abbreviations may not be known by those that are not technical experts in the field. These include: GCN, GAT, GIN, GAT-GCN, CNN, SHAP, GINE-based, MLP, GELU, GNN.

Response. We defined the technical terms at first use, including “graph-neural-network (GNN) setting,” and standardised GINE as “GINE-based graph encoder.”

Revision in manuscript: Introduction; Related Work; Methods.

R3.10. Minor comment 2: atom-by-residue terminology

Reviewer comment (full text). *Minor comment 2.* On page 1 (3rd line from the bottom), the authors mention biologically meaningful resolution through atom-by-residue interaction maps. Atom-by-residue is not standard terminology and is a bit confusing in the absence of proper context. Maybe the authors can write this in a more informative way, e.g., interaction maps that show which ligand atoms interact with residues of the protein.

Response. We replaced it with “atom–residue attention matrix” or “model-native attribution map” and explicitly state that it is not a contact map.

Revision in manuscript: Abstract; Introduction; Methods; figure captions.

R3.11. Minor comment 3: original Figure 2

Reviewer comment (full text). *Minor comment 3. Figure 2:* The authors should refer to the two sub-panel figures by their panel letters (e.g., Ablation study results. Panel A summarizes random-split and cold-target metrics as a heatmap, and Panel B reports the AUROC decrease relative to the full model). Also, the figure in Panel B appears to be improperly formatted for the page. The panel label B overlaps with the figure title, the figure legend (Random AUROC and Cold-target AUROC) overlaps with the lowermost bar graph, and the x-axis labels are overlapping with one another, making it difficult to read the numerical delta values. The authors are requested to revise this figure for clarity.

Response. We restored and redrew the ablation figure. Panel labels are now explicit, the Panel B legend sits below the bars, the delta labels no longer overlap, and all axis labels are readable at the manuscript’s single-column width. The corresponding component-removal values are provided in Table 2.

Revision in manuscript: Results, “Ablation Study”; Table 2 and Figure 2.

R3.12. Minor comment 4: original Figure 3

Reviewer comment (full text). *Minor comment 4. Figure 3:* The authors should refer to the two sub-panel figures by their panel letters (e.g., Grouped bar charts display A) AUROC, AUPRC, and B) F1, Precision, and Recall). For the figure in Panel A, the error bars overlap with the numerical score labels, and for the figure in Panel B, the numerical score labels either overlap with each other (in the Recall case), or are shifted leftwards relative to the bar they represent. The authors are requested to revise this figure for clarity.

Response. We redrew the figure as threshold-free AUROC/AUPRC bars from the canonical current-release entity-split reconstruction. The problematic threshold-dependent labels were removed; validation-threshold F1, precision, and recall are reported in Supporting Information Table S5, and corresponding strict-split values are reported in Table S7.

Revision in manuscript: Revised Fig. 3 and caption.

R3.13. Minor comment 5: original Figure 4

Reviewer comment (full text). *Minor comment 5. Figure 4:* As with the other figures, the authors should make sure to refer to the sub-panels by their labels (e.g., A) validation AUROC over training epochs. B) training loss).

Response. Figure 5 now identifies Panel A as validation AUROC and Panel B as training loss. Each line is a single contiguous archived trace selected by agreement between its peak validation AUROC and the saved checkpoint; no fitted or interpolated trend is displayed. The caption limits interpretation to observed single-seed trajectories.

Revision in manuscript: Revised Figure 5 and caption.

R3.14. Minor comment 6: original Figure 5

Reviewer comment (full text). Minor comment 6. Figure 5: Panel A: The figure legend overlaps with the histogram plot. Shifting the figure legend to the right side would aid with clarity.

Response. We rebuilt Figure 6 from the archived 1,301,380 pair-normalised residue scores. Panel A is their logarithmic-scale distribution and Panel B is the cumulative attribution mass after sorting the same values; the figure has no overlapping legend. The caption defines the attribution-concentration statistic and does not present it as structural validation.

Revision in manuscript: Revised Figure 6 and caption.

R3.15. Minor comment 7: original Figure 6

Reviewer comment (full text). Minor comment 7. Figure 6: As only one figure is being shown, a subpanel label (A) is not needed for this figure.

Response. The single-panel ABL1–D0017 cross-attention heatmap from the original submission was removed after the input audit showed that the nominal ABL1 variant analysis was not supportable. It therefore no longer carries a panel label or a caption in the revised manuscript.

Revision in manuscript: Removed the original single-panel ABL1 heatmap as part of the input-audit revision.

R3.16. Minor comment 8: original Figure 8

Reviewer comment (full text). Minor comment 8. Figure 8: The authors should refer to the two sub-panel figures by their panel letters (e.g., Attention conservation across ABL1 mutants. Panel A: pairwise correlation matrix of full residue-attention profiles for compound D0017 across five ABL1 variants. Panel B: attention assigned to the three most conserved hotspot residues). Also, the figure in Panel B appears to be improperly formatted for the page. The figure legend (V104, A648, S199) overlaps with the rightmost bar graph for E255K, and the numerical labels for the normalized attention are overlapping with one another, making it difficult to read the values. The authors are requested to revise this figure for clarity.

Response. Following the input audit described above, we withdrew the entire mutant-comparison figure rather than repair its layout. The figure used nominal ABL1 variant labels whose model-input sequences are identical, so neither its correlation matrix nor its bar chart can support a variant-specific analysis.

Revision in manuscript: Removed the former mutant-comparison figure; added the transparent ABL1 input-audit limitation and Supporting Information Table S9.

R3.17. Minor comment 9: original Figure 9

Reviewer comment (full text). Minor comment 9. Figure 9: As with the other figures, the authors should make sure to refer to the sub-panels by their panel letters (e.g., Panel A reproduces the atom-level Grad-CAM map; Panel B ranks functional-group importance). Also, for Panel A, the authors do not indicate what the importance thresholds are for the functional group colors in the Grad-CAM analysis. The authors should include some information about what values the different color intensities (yellow, orange, red, maroon) correspond to.

Response. We removed the former ABL1(F317I) Grad-CAM panel because the nominal variant does not have a variant-specific model sequence. The retained non-mutant EGFR functional-group table reports its saved normalised values directly, avoiding a colour-scale interpretation that could be mistaken for physical contacts.

Revision in manuscript: Removed the former ABL1(F317I) Grad-CAM figure; Results, “Case Study: Functional-Group Attribution”.

R3.18. Minor comment 10: original Figure 10

Reviewer comment (full text). *Minor comment 10. Figure 10: As with the other figures, the authors should make sure to refer to the sub-panels by their panel letters (e.g., Functional-group importance for ABL1 (D0017), EGFR (D0014), and BRAF (D0023) is displayed as a comparative map derived from Grad-CAM outputs (Panel A), accompanied by a bar chart of binding probabilities (Panel B)).*

Response. We removed the three-case figure because it included the invalid nominal ABL1 variant input. Its non-mutant EGFR/BRAF numerical summaries are now given in a compact Results table, which reports classifier scores and the largest saved functional-group attribution without implying a physical interaction mechanism.

Revision in manuscript: Results, “Cross-Case Functional-Group Attribution Variation”.

R3.19. Minor comment 11: original Figure 11

Reviewer comment (full text). *Minor comment 11. Figure 11: As with the other figures, the authors should make sure to refer to the sub-panels by their labels (e.g., Panel A: scatter plot of experimental Kd versus hotspot count. Panel B: tabulated hotspot counts and binding probabilities for each case). Also, for the figure in Panel A, there is only one x-axis label for Experimental Kd (nM). It could be assumed that the scale is logarithmic, but additional x-axis labels should be added to avoid any ambiguity in interpretation. The label for BRAF-V600E also overlaps with the color bar on the right side. Please adjust the positioning of this label for readability.*

Response. We removed the hotspot-count figure. It pooled a small set of selected cases that included nominal ABL1 variants with identical model-input sequences, and its thresholded count cannot establish a physical or causal relationship to affinity. Removing the panel is more transparent than a cosmetic repair.

Revision in manuscript: Removed the former hotspot-count analysis and figure.

R3.20. Minor comment 12: original Figure 12

Reviewer comment (full text). *Minor comment 12. Figure 12: This figure does not show the residue-level cross-attention heat map or ranked hotspot list, as claimed in the figure legend. The authors should provide a screenshot that accurately depicts what they claim in the figure legend. Also, the authors should indicate that their screenshot uses the protein sequence of Human Aurora Kinase C (UniProt ID: Q9UQB9) so that readers are able to reproduce the output and validate it for themselves.*

Response. The caption now accurately identifies the custom-prediction input/result state for human Aurora kinase C (Q9UQB9), including entered SMILES, protein sequence, and high-affinity-class classifier score (sigmoid output). The lower portion of the screenshot also shows the generated atom-residue attention map and the “Top 10 Residue Attributions” chart. The caption states that these are model-native attribution views rather than physical-contact or hotspot labels. Revised Fig. 7 now presents Panel A as the full interface and Panel B as an enlarged crop of the attribution region so that the heatmap and ranked-residue chart are legible in the manuscript.

Revision in manuscript: Figure 7 and caption.

R3.21. Remarks on code availability

Reviewer comment (full text). *The authors have released a public version of the code on GitHub (<https://github.com/Liubo-520/bio-intelligence>) with instructions for installation, training, and replication of results. They have provided checkpoint files so that users can directly run the model without having to perform the training step.*

Response. We thank the reviewer for noting the public repository. The revised manuscript now provides direct access to the BioInteract source repository at <https://github.com/Liubo-520/bio-intelligence>, which contains the code, pretrained weights, configuration files, figure-generation scripts, and reproduction instructions, and to the immutable Zenodo archive, <https://doi.org/10.5281/zenodo.21437753>, which preserves the release snapshot, canonical prediction artifacts, and provenance hashes. It also links to the strict BindingDB aggregate manifests, curation/evaluation code, and audit report while excluding the raw third-party archive and derived ESM cache. The manuscript also links to the public interactive interface at <https://huggingface.co/spaces/AI4deeperScience/BioInteract> for the custom-prediction workflow.

Revision in manuscript: Data Availability; Code Availability.

Summary of substantive changes

- Added full-length target-identity audit, exact-sequence-grouped cold-target and exact-sequence-grouped cold-both tests, and identity-stratified strict-test metrics.
- Added leakage-aware diagnostic controls that expose substantial drug-side signal in the Davis exact-sequence-grouped cold-target setting.
- Added a pre-specified, strict double-novel BindingDB K_d -only external evaluation using a frozen Davis checkpoint and threshold; the modest result is reported as limited transfer rather than broad DTI generalisation.
- Rebuilt the original entity-split metrics from released checkpoints and inputs under a frozen validation-threshold protocol and used the updated BioInteract values in the restored baseline-comparison table and figure.
- Restored the original numerical cross-model comparison and component-removal rows; the BioInteract full-model row uses canonical current release metrics, Figure 2 is regenerated from the displayed rows, and the component differences are explicitly limited to descriptive rather than controlled matched-retraining evidence. The baseline comparison remains benchmark context rather than a causal architecture comparison.
- Completed attention-sparsity, bootstrap, sequence-similarity, fingerprint, software-version, and Grad-CAM reproducibility specifications.
- Restricted novelty, baseline, generalisation, binary-task, and interpretability claims to what the reported evidence supports.
- Withdrew all nominal-ABL1-variant comparisons after auditing identical model-input sequences; retained only non-mutant EGFR/BRAF functional-group attribution summaries.
- Added response-letter evidence tables and revised figure panels.
- Added direct GitHub, Zenodo DOI, and Hugging Face links for the code, immutable release, model resources, and interactive custom-prediction interface.

We thank the Editor and Reviewers again for improving the precision, scope, and transparency of the manuscript.